

■ EDUCATION

University of California, Irvine

B.S. in Physics · GPA 3.81/4.00 · Dean's Honor List, six quarters

Sep. 2023 – Jun. 2027 (expected)

Irvine, CA

- **Graduate coursework:** Quantum Field Theory, Quantum Mechanics, Group Theory, Machine Learning & Statistics.

■ RESEARCH INTERESTS

High Energy Physics (HEP) · AI/ML for HEP · BSM phenomenology · Dark-matter searches

■ PUBLICATIONS & MANUSCRIPTS

PARTICLE PHYSICS

- [1] Max H. Fieg, Patrick J. Fox, **Jinbo Zhang**, Aishik Ghosh, Virat Varada, and Daniel Whiteson. "Time-dependent signals of new physics at the LHC." [arXiv:2605.11071](#) (2026).
UNDER REVIEW AT *Physical Review D* (REVISED MANUSCRIPT)
- [2] **Jinbo Zhang**, Bertrand Echenard, and David G. Hitlin. "A Muon Missing-Momentum Experiment." LDMX Collaboration internal note (2026).
FIRST AUTHOR · CALTECH SURF FINAL REPORT, POSTED TO THE LDMX DOCUMENT DATABASE · DRAFT AVAILABLE ON REQUEST
- [3] **Jinbo Zhang**, Jan Eysermans, Sara Aumiller, and Christoph Paus. "Measurement of $H \rightarrow WW^*$ in the $q\bar{q} \ell\nu q\bar{q}'$ final state at FCC-ee." (2026).
FIRST AUTHOR · MANUSCRIPT COMPLETE; TO BE RELEASED AS AN FCC-EE NOTE · DRAFT AVAILABLE ON REQUEST

CO-AUTHORED PUBLICATIONS AND PROCEEDINGS

- [4] Euclid Collaboration: X. Xu, R. Chen, T. Li, A. R. Cooray, . . . , **J. Zhang**, et al. "Euclid Quick Data Release (Q1). AgileLens: A scalable CNN-based pipeline for strong gravitational lens identification." [arXiv:2604.06648](#) (2026).
- [5] Christopher M. Gonzalez, Yu-Hsun Chen, Christopher Effarah, **Jinbo Zhang**, et al. "Gate-tunable two-dimensional ITO epsilon-near-zero materials fabricated by liquid metal printing." *Proc. SPIE* PC13579, PC135791B (2025).
- [6] Ho Wai (Howard) Lee, Quynh Dang, David Dang, et al., incl. **Jinbo Zhang**. "Active and nonlinear epsilon-near-zero photonics." *Proc. SPIE* PC13892 (2026).
- [7] R. M. Valencia, G. Xing, A. Subbaraman, **Jinbo Zhang**, A. Cooray, and T. Bakx. "The redshift distribution and physical properties of Herschel-selected ultrared dusty star-forming galaxies with ALMA Band-3 spectral scans." IN PREPARATION

■ TALKS & POSTERS

- **Jinbo Zhang**. "A Muon Missing-Momentum Experiment." **Invited talk**, LDMX Collaboration meeting (Fall 2026).
- **Jinbo Zhang**. "A Muon Missing-Momentum Experiment." Final research talk, Caltech SURF Program (Aug. 2026).
- **Jinbo Zhang**. "Time-Dependent Signals of New Physics at the LHC." Poster, UCI Undergraduate Research Symposium (May 2026).
- Yu-Hsun Chen, **Jinbo Zhang**, et al. "Liquid Metal Printed 2D ITO for Nanophotonic Applications." Poster, UCI Undergraduate Research Symposium (May 2025).

■ RESEARCH EXPERIENCE

Muon Missing-Momentum Search for Dark Matter | Caltech SURF

Jun. 2026 – Present

Mentors: Prof. Bertrand Echenard and Prof. David Hitlin

California Institute of Technology

- Built, in an eight-week program, the complete simulation and analysis chain for an LDMX-style muon-beam missing-momentum experiment: a full Geant4 model (40-layer LYSO active target with per-crystal readout, electromagnetic and hadronic calorimeters) and a 10^8 -muon background campaign on a Slurm cluster. **First author** of the resulting manuscript.
- Mapped the rare-background landscape of the experiment at the one-in- 10^8 level, identifying muon bremsstrahlung with photonuclear interactions and hard muon-nucleus scattering with escaping neutrons as the irreducible processes, and tracing every surviving event topology to its physical mechanism.
- Designed a 625-node graph neural network over the full detector and benchmarked it against calorimetric vetoes, BDTs, and transformers, establishing a **background-free working point at 10^8 muons with $\sim 90\%$ signal efficiency** under a preregistered model-selection and validation protocol. Invited to present the results to the LDMX Collaboration.

Precision Higgs Physics at FCC-ee: $\sigma(\text{ZH}) \times \mathcal{B}(\text{H} \rightarrow \text{WW}^*)$

Mentor: Prof. Christoph Paus

Jan. 2026 – Present
Massachusetts Institute of Technology

- **Conceived and led, as first author**, a complete projected measurement of $\sigma(\text{ZH}) \times \mathcal{B}(\text{H} \rightarrow \text{WW}^*)$ in the semileptonic $q\bar{q} \ell \nu q\bar{q}'$ final state at $\sqrt{s} = 240$ GeV (10.8 ab^{-1} , IDEA detector), owning the full chain from event selection to final fit and processing over 10^9 fully simulated events across 32 signal and background processes in FCCAnalyses.
- Achieved a **sub-percent projected precision on the signal strength** ($\delta\mu/\mu = 0.595\%$) from a 20-bin profile-likelihood fit in pyhf, hardened with Asimov closure, likelihood scans, cross-validation, and systematic selection scans.
- Invented a Z-priority jet-pairing algorithm that uses the on-shell Z to resolve the off-shell W^* topology, and trained an XGBoost classifier on 26 kinematic observables (five-fold out-of-fold AUC = 0.975). [analysis code]

Time-Dependent Searches for New Physics at the LHC

Mentor: Prof. Daniel Whiteson

May 2025 – May 2026
University of California, Irvine

- Co-developed a new resonance-search paradigm that adds event time as an analysis dimension, opening sensitivity to temporally modulated signals that conventional mass-only searches cannot see. **Co-author of the resulting paper, now under review at Physical Review D.**
- Extended CATHODE weakly supervised anomaly detection from one to two dimensions, learning background distributions jointly in invariant-mass and time sidebands for fully data-driven searches.
- Built the complete simulation-to-limits pipeline (MadGraph5, Pythia8, Delphes, classifier training, sideband-closure diagnostics, expected upper limits) and produced the sensitivity comparisons between mass-only and mass-plus-time strategies that underpin the paper. [code]

■ ADDITIONAL RESEARCH EXPERIENCE

High-Redshift Galaxy Evolution

Mentor: Prof. Asantha Cooray

Dec. 2024 – Sep. 2025
University of California, Irvine

- Analyzed Spitzer and ALMA photometry of Herschel-selected ultrared dusty star-forming galaxies and built automated Python/CARTA pipelines for spectral-energy-distribution fitting; co-author on the resulting manuscript.

Epsilon-Near-Zero Nanophotonics

Mentor: Prof. Howard Lee

Feb. 2024 – Sep. 2025
University of California, Irvine

- Fabricated gate-tunable two-dimensional ITO films by liquid-metal printing and characterized their nonlinear optical response; co-author on two SPIE proceedings and a symposium poster.

■ TECHNICAL SKILLS

HEP simulation & analysis

Geant4, MadGraph5, Pythia8, Delphes, FCCAnalyses, ROOT (PyROOT, RDataFrame), pyhf

Machine learning

PyTorch, XGBoost, graph neural networks, transformers, weakly supervised anomaly detection (CATHODE)

Programming & computing

Python, C++, NumPy/Pandas, Slurm-based HPC, Git

Languages

Chinese (native), English (professional)

■ TEACHING & LEADERSHIP

Learning Assistant, Classical Mechanics and Electromagnetism

Department of Physics and Astronomy

Jan. 2025 – Present
University of California, Irvine

- Lead weekly discussion and laboratory sections for upper-division mechanics and electromagnetism, guiding students through experimental design, uncertainty analysis, and quantitative problem solving.

Webmaster, Society of Physics Students at UCI

Executive board

2026 – 2027
University of California, Irvine

- Maintain the chapter website and publish events, resources, and outreach materials for the undergraduate physics community.

■ HONORS & AWARDS

- Caltech Summer Undergraduate Research Fellowship (SURF), 2026, \$8,110 award.
- UCI Undergraduate Research Opportunities Program (UROP) Award, two consecutive years (2024, 2025).
- Dean's Honor List, University of California, Irvine, six quarters (2024–2026).